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PRACTICAL 6: LAN CONSTRUCTION AND SWITCH CONFIGURATION

Objective

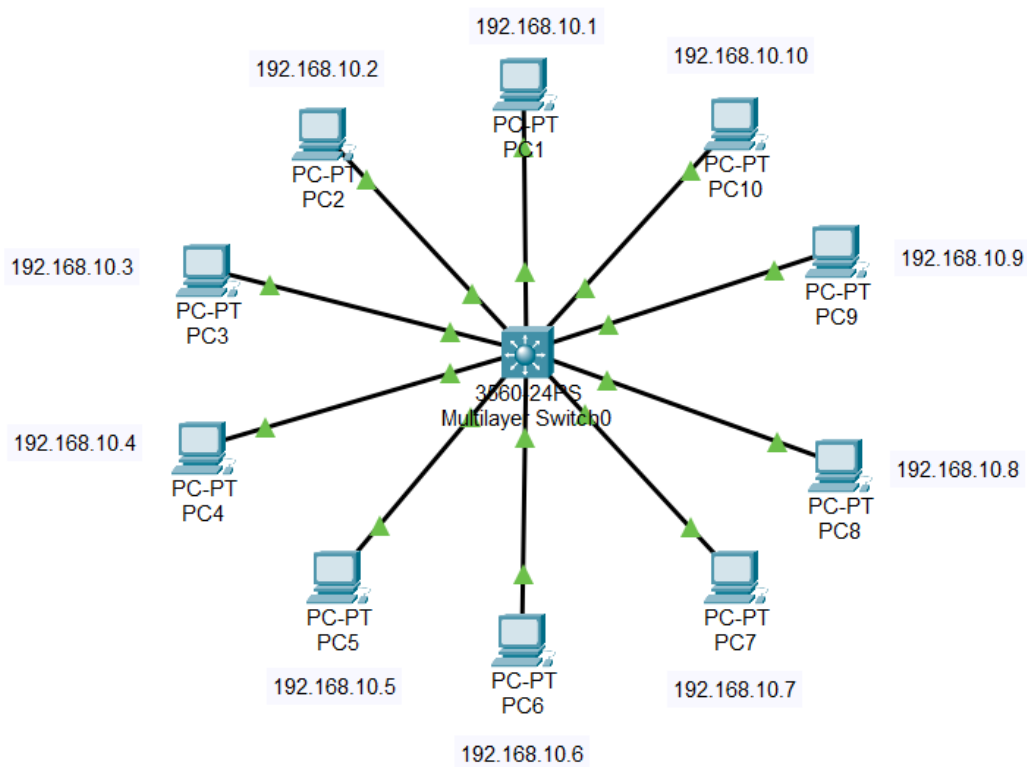
Build a LAN using a Cisco Catalyst switch.

Equipment

- 10 PCs
- Cisco Catalyst 3560 Switch
- Patch Panel
- Fly Leads

Activities

1. Connect all 10 PCs.



2. Configure static IP addresses.

PC1

☐ DHCP	☒ Static
IPv4 Address	192.168.10.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC2

☐ DHCP	☒ Static
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC3

☐ DHCP	☒ Static
IPv4 Address	192.168.10.3
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC4

☐ DHCP	☒ Static
IPv4 Address	192.168.10.4
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC5

☐ DHCP	☒ Static
IPv4 Address	192.168.10.5
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC6

☐ DHCP	☒ Static
IPv4 Address	192.168.10.6
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC7

☐ DHCP	☒ Static
IPv4 Address	192.168.10.7
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC8

☐ DHCP	☒ Static
IPv4 Address	192.168.10.8
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC9

☐ DHCP	☒ Static
IPv4 Address	192.168.10.9
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

PC10

☐ DHCP	☒ Static
IPv4 Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

3. Configure switch management interface.

```
Switch>ENABLE
Switch#confi ter
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.10.254 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#exit
Switch(config)#copy running-config startup-config
^
% Invalid input detected at '^' marker.

Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

4. Verify connectivity.

Ping from PC1 to every device:

Switch

```
C:\>PING 192.168.10.254

Pinging 192.168.10.254 with 32 bytes of data:

Reply from 192.168.10.254: bytes=32 time<1ms TTL=255
Reply from 192.168.10.254: bytes=32 time<1ms TTL=255
Reply from 192.168.10.254: bytes=32 time<1ms TTL=255
Reply from 192.168.10.254: bytes=32 time<1ms TTL=255
```

PC2

```
C:\>PING 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
```

PC3

```
C:\>PING 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

PC4

```
C:\>PING 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time=4ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
```

PC5

```
C:\>PING 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
```

PC6

```
C:\>PING 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
Reply from 192.168.10.6: bytes=32 time<1ms TTL=128
```

PC7

```
C:\>PING 192.168.10.7

Pinging 192.168.10.7 with 32 bytes of data:

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
```

PC8

```
C:\>PING 192.168.10.8

Pinging 192.168.10.8 with 32 bytes of data:

Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
Reply from 192.168.10.8: bytes=32 time<1ms TTL=128
```

PC9

```
C:\>PING 192.168.10.9

Pinging 192.168.10.9 with 32 bytes of data:

Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
```

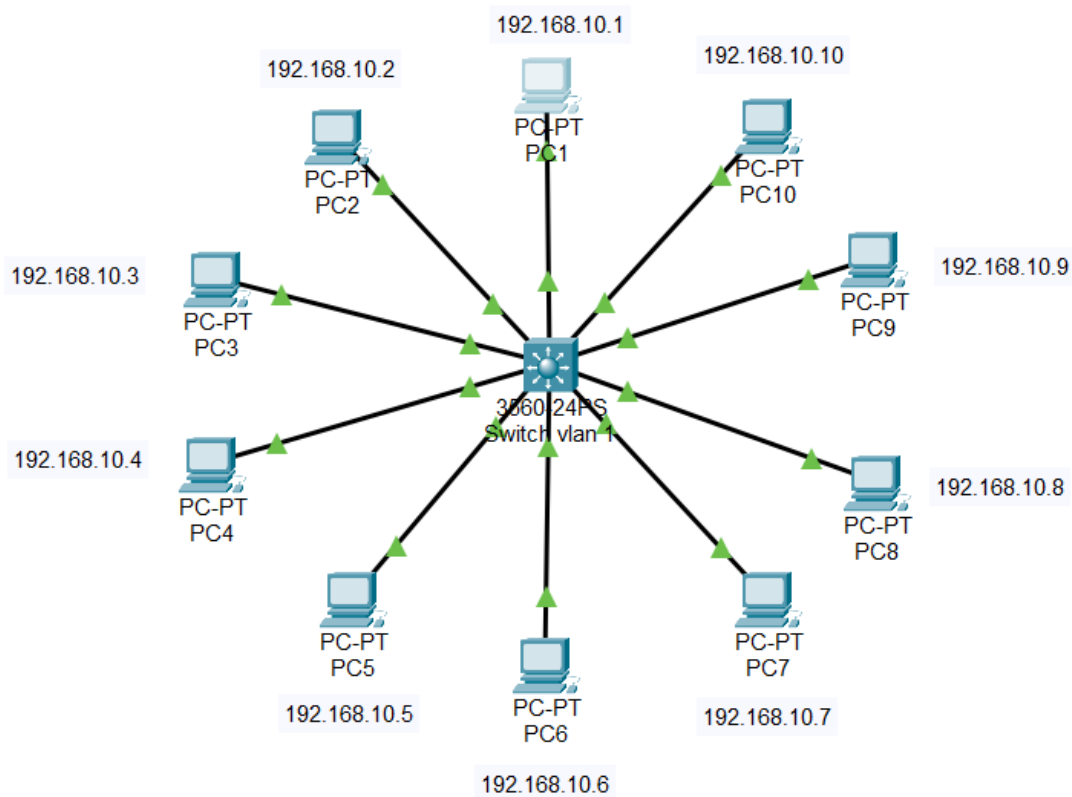
PC10

```
C:\>PING 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128
```

5. Create a network diagram.



IP Addressing Scheme	Device	IP Address
PC1		192.168.10.1
PC2		192.168.10.2
PC3		192.168.10.3
PC4		192.168.10.4
PC5		192.168.10.5
PC6		192.168.10.6
PC7		192.168.10.7
PC8		192.168.10.8
PC9		192.168.10.9
PC10		192.168.10.10
Switch VLAN 1		192.168.10.254

Verification Commands

1 show running-config

```
Switch#show running-config
Building configuration...

Current configuration : 1166 bytes
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
```

2 show vlan brief

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

3 show mac address-table

```
Switch#show mac address-table
Mac Address Table
```

Vlan	Mac Address	Type	Ports
1	0001.4388.2643	DYNAMIC	Fa0/8
1	0001.637c.9411	DYNAMIC	Fa0/2
1	0002.16ca.a52b	DYNAMIC	Fa0/10
1	0003.e4b9.0359	DYNAMIC	Fa0/3
1	000c.cf8c.b724	DYNAMIC	Fa0/9
1	0060.5cb8.7404	DYNAMIC	Fa0/6
1	00d0.9713.5ddc	DYNAMIC	Fa0/1
1	00d0.9734.a240	DYNAMIC	Fa0/5
1	00d0.ff33.b380	DYNAMIC	Fa0/4
1	00e0.f7cc.a93d	DYNAMIC	Fa0/7

```
Switch#
```

Deliverable

Operational LAN and topology diagram.

Criteria	Marks
Physical installation	3
Switch configuration	4
IP configuration	4
Connectivity testing	4
Total	15